

End-to-End Retail Performance and Behavioral Analytics

1. Project Overview

This project focuses on analyzing retail sales, customer behavior, product performance, store operations, and return patterns using Python, SQL, and Power BI. The objective is to transform raw retail data into actionable business insights through data cleaning, analysis, and interactive dashboards.

Tools Used:

- Python (Pandas, NumPy, Matplotlib)
 - SQL Server
 - Power BI
 - Excel/CSV Files
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2. Business Objective

The primary objectives of this project are:

- Analyze overall sales performance and revenue trends.
 - Understand customer demographics and purchasing behavior.
 - Evaluate product and category profitability.
 - Monitor return patterns and customer return behavior.
 - Provide data-driven recommendations for improving business performance.
 - Build interactive dashboards for decision-making.
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3. Dataset Description

The project uses five datasets:

Sales Data

Contains transaction-level sales information including:

- Order ID
- Order Date
- Customer ID
- Product ID
- Store ID
- Sales Channel
- Quantity
- Unit Price

- Discount Percentage
- Total Amount

Customer Data

Contains customer demographic information:

- Customer ID
- Gender
- Age
- Signup Date
- Region

Product Data

Contains product-related information:

- Product Name
- Category
- Brand
- Cost Price
- Unit Price
- Margin Percentage

Store Data

Contains store details:

- Store Name
- Store Type
- Region
- City
- Operating Cost

Returns Data

Contains returned order information:

- Return ID
- Order ID
- Return Date
- Return Reason

4. Data Cleaning Summary

Data cleaning was performed using Python.

Cleaning Activities

- Removed duplicate records.
- Handled missing values.
- Standardized column names.

- Converted date columns to datetime format.
- Corrected data types.
- Removed unnecessary spaces and inconsistencies.
- Created cleaned datasets:
 - cleaned_sales.csv
 - cleaned_customers.csv
 - cleaned_products.csv
 - cleaned_stores.csv
 - cleaned_returns.csv

Feature Engineering

Created:

- Profit calculations
 - Revenue metrics
 - Customer Age Groups
 - Date hierarchy columns
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5. SQL Analysis

SQL Server was used to build the retail analytics database, store the cleaned datasets, enforce data integrity, and perform business analysis using SQL queries.

Database Creation

The following relational tables were created:

- Sales
- Customer
- Products
- Stores
- Returns_Data

Data Loading

- Imported cleaned CSV files into SQL Server using the **BULK INSERT** command.
- Verified successful data loading by checking row counts and table contents.
- Standardized data types for all columns.

Data Validation

- Checked for duplicate records in all tables.
- Removed duplicate records where necessary.
- Validated NULL values and corrected data inconsistencies.
- Verified record counts after data loading.

Database Design

- Defined appropriate data types for all attributes.

- Created Primary Keys for:
 - customer
 - products
 - stores
 - returns_data
 - sales (using order_id after data validation)
- Established Foreign Key relationships between related tables.
- Generated an Entity Relationship (ER) Diagram representing the database schema.

Business Metrics

Calculated derived business metrics including:

- Profit
- Total Sales Amount
- Margin Percentage
- Order Year
- Order Month

Key SQL Queries Performed

1. Total Revenue

- Calculated total revenue generated from all sales transactions.

2. Total Profit

- Calculated overall profit generated by the business.

3. Top 5 Selling Products

- Identified the highest-selling products based on quantity sold.

4. Monthly Sales Trend

- Analyzed monthly revenue and sales performance.

5. Sales by Channel

- Compared Online and In-Store sales performance.

6. Customer Analysis

- Analyzed customer distribution across different regions.

7. Product Category Performance

- Evaluated revenue and profit generated by each product category.

8. Store Performance

- Compared sales performance across different stores and regions.

9. Return Analysis

- Analyzed returned orders and identified the most common return reasons.

10. Profitability Analysis

- Measured profit generated by individual products and product categories.

Database Optimization

- Created indexes on key columns to improve query performance.
 - Verified indexes using SQL Server system procedures.
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6. Dashboard Overview

Interactive dashboards were developed using **Microsoft Power BI** to visualize key business metrics, monitor retail performance, and support data-driven decision-making. The dashboards include interactive filters, KPI cards, drill-down analysis, and dynamic visualizations.

Dashboard 1: Sales Performance Overview

KPIs

- Total Revenue
- Total Orders
- Total Quantity Sold

Charts

- Monthly Revenue Trend
 - Sales by Channel (Online vs In-Store)
 - Sales by Category
 - Top Products by Quantity Sold
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Dashboard 2: Customer Segmentation & Insights

KPIs

- Total Revenue
- Total Customers
- Average Customer Age

Charts

- Revenue by Age Group
- Customer Age Distribution
- Top Customers by Revenue
- Customer Count by Region

Dashboard 3: Product & Category Analysis

KPIs

- Total Profit
- Total Products
- Total Quantity Sold

Charts

- Profit by Category (Bar Chart)
- Profit by Category (Donut Chart)
- Top Products by Revenue
- Sales Quantity by Product and Region

Dashboard 4: Return Behavior Patterns

KPIs

- Total Returns
- Return Rate
- Total Orders

Charts

- Return Trend
- Return Reasons Distribution
- Return Rate by Product Category
- Quarterly Return Analysis

Dashboard 5: Stores Analysis

KPIs

- Total Profit
- Total Stores
- Total Quantity Sold

Charts

- Top 5 Highest Profit Stores
- Stores with Highest Sales

Store Profitability (Profit vs Operating Cost Scatter Plot)

Dashboard Features

- Interactive slicers for Month, Year, Region, Category, Product, Store, Sales Channel, Gender, Age Group, and Return Reason.
 - Dynamic KPI cards that update automatically based on selected filters.
 - Reset Filters button for improved dashboard navigation.
 - Cross-filtering and drill-down capabilities for detailed analysis.
 - Consistent visual theme and layout across all dashboards for enhanced user experience.
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7. Key Findings

Sales Insights

- Generated **\$818.13K** in total revenue from approximately **3,000 orders**.
- In-Store sales contributed more revenue than Online sales.
- Personal Care was the highest-performing product category.
- A few top-selling products generated a significant share of total revenue.

Customer Insights

- The business served **782 customers** with an average age of **44 years**.
- Customers aged **55+** generated the highest revenue.
- Customer distribution was balanced across all regions.
- High-value customers contributed significantly to overall sales.

Product Insights

- Personal Care generated the highest profit.
- Product profitability varied across different categories.
- Top-selling products contributed a large share of total revenue.

Store Insights

- Total profit across **60 stores** was approximately **\$243.75K**.
- Store performance varied by location.
- Some stores achieved high profits while maintaining lower operating costs.

Return Insights

- A total of **193 returns** were recorded.
 - Defective products were the most common return reason.
 - Return trends highlighted opportunities to improve product quality and operations.
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8. Business Recommendations

Sales

- Focus promotions on top-selling products.
- Increase online marketing to improve digital sales.

- Run seasonal campaigns during low-sales periods.

Customers

- Introduce loyalty programs for high-value customers.
- Develop personalized marketing campaigns.
- Strengthen customer retention strategies.

Products

- Maintain adequate stock for high-demand products.
- Review pricing and discount strategies regularly.
- Improve or replace low-performing products.

Stores

- Reduce operating costs in underperforming stores.
- Apply best practices from top-performing stores.
- Monitor store performance regularly.

Returns

- Improve product quality control.
- Strengthen packaging and delivery processes.
- Monitor return reasons to reduce future returns.

9. Conclusion

This project successfully analyzed retail sales data using **Python, SQL Server, and Power BI**. Python was used for data cleaning and preprocessing, SQL Server for database management and business analysis, and Power BI for creating interactive dashboards.

The dashboards provided valuable insights into **sales performance, customer behavior, product profitability, store performance, and return patterns**. The findings and recommendations can help businesses make informed decisions, improve operational efficiency, increase profitability, and enhance customer satisfaction.